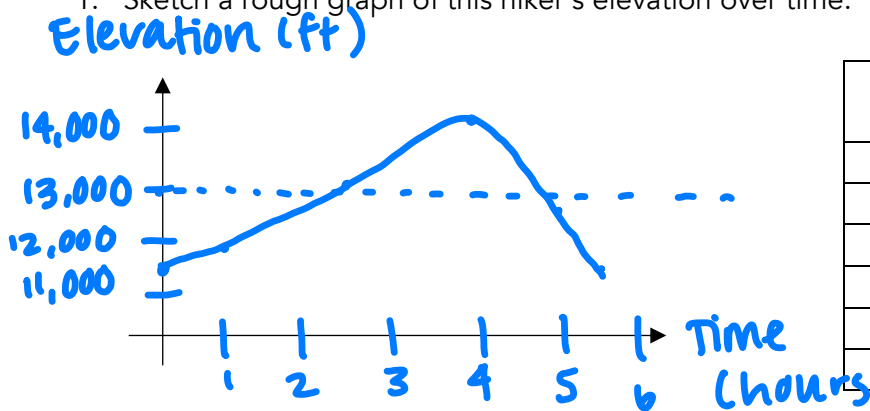


How Long to Reach the Summit?



Mount Bierstadt Trail is a popular 6.8 mile out-and-back trail in Silver Plume, Colorado. It is one of Colorado's iconic 14-ers, with a peak of 14,066 feet. The table below gives the elevation of a hiker climbing Mt. Bierstadt at selected times on his trip. Time is given in hours and elevation is given in feet.

1. Sketch a rough graph of this hiker's elevation over time.



t **E(t)**

Time (hours)	Elevation (above sea level)
0	11,629 feet
1	11,900 feet
2.5	13,060 feet
4	14,066 feet
5	12,619 feet
5.5	11,629 feet

2. Can the hiker's elevation be considered a function? Why or why not?

yes, at each moment in time the hiker is at only one elevation.

3. Explain why each elevation (except the summit) is reached at least twice on the hiker's journey.

Each elevation is reached once on the way up the mountain and once on the way down.

4. Suppose that we wanted to find the *inverse*: to figure out the time at which the hiker reached a certain elevation.

- a. Identify the input and output of this inverse relation.

Input: Elevation Output: Time

- b. Estimate the time at which the hiker was at an elevation of 10,500 feet or explain why this is not possible.

Not possible because the hiker started & ended above 10,500.

- c. Explain how you can use the graph to estimate at what time the hiker was half-way up the mountain (12,978 feet).

Sketch a line at 12,978 ft and see at what time the hiker reached that elevation ≈ 2.4 hours and 4.8 hrs.

Is the inverse relation a function? How do you know?

No, because a single elevation outputs multiple times.

Time...
Math Medic
Elevation

E(t) has repeated y-values; not one-to-one

10,500 is not in range of *E(t)* so not in domain of inverse

Repeated y-values on original repeated x-values on inverse $\Rightarrow$

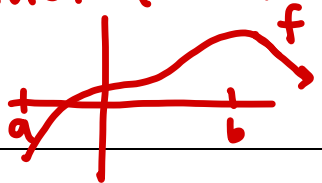
Lesson 1.10 —Graphs of Inverse Functions

QuickNotes

In order for $f(x)$ to have an inverse that's also a function, $f(x)$ must be one-to-one (no repeated y-values)

This can be checked w/ a horizontal line test.

A function can be one-to-one on a restricted interval of the domain.



f is one-to-one on (a,b) but not for all $\mathbb{R}$

Domain of f is range of f^{-1}
Range of f is domain of f^{-1}

Check Your Understanding

- The function $H(t)$ gives the temperature of a town, in °Fahrenheit, t days into the calendar year.

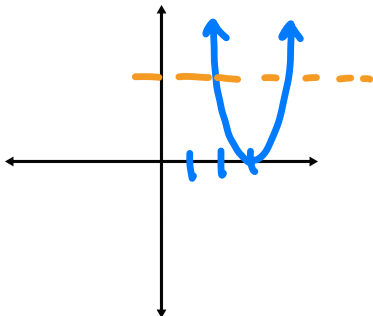
- Is $H(t)$ a one-to-one function? How do you know?

No, multiple days of the year could have the same temperature.

- Explain what this tells you about whether or not H^{-1} is a function.

Since the range of H is the domain of H^{-1} we would have identical inputs going to different outputs so H^{-1} is not a function.

- Sketch the graph of $f(x) = (x - 3)^2$. Is the inverse of f a function? Support your answer with information from the graph. If not, how could you restrict the domain of f so that the inverse is a function?



No, the inverse is not a function because f fails the HLT, which means f^{-1} fails the VLT

We could restrict domain to $(-\infty, 3]$ or $[3, \infty)$

- Find the inverse of $y = \sqrt{x - 4}$ and state its domain and range.

Original $y = \sqrt{x - 4}$

Inverse $x^2 + 4 = y$

D: $x \geq 0$

R: $y \geq 4$

Domain of original: $x \geq 4$

Range of inverse: $y \geq 4$

Range of original: $y \geq 0$

Domain of inverse: $x \geq 0$